

CS & DA



2024



Warehousing

DPP-02 Discussion Notes

By-Dr. Astha Singh



#Q. Which of the following is not a data cleaning technique?

A Outlier detection and removal

B Imputation of missing values

C Data transformation

D Data segmentation

Related to data partitioning rather than cleaning

#Q. In data cleaning, what is outlier detection used for?

A

To add noise to the data

B

To remove incorrect or irrelevant data points

C

To balance the data distribution

D

To normalize the data range

#Q. Which of the following is an example of data smoothing technique?

A

Principal Component Analysis (PCA)

B

~~Moving Average~~

— Reduce noise
averaging neighbouring data points

C

Logistic Regression

D

K-means clustering

#Q. What does the term "data deduplication" refer to in data cleaning?

- A** Adding duplicate entries to the dataset
- B** ~~Detecting and removing duplicate entries from the dataset~~
- C** Splitting the dataset into smaller segments
- D** Converting categorical data to numerical data

#Q. Which of the following is a disadvantage of data smoothing techniques?

- A** They ~~reduce the~~ interpretability of data patterns
- B** They increase the likelihood of overfitting
- C** They can only handle numerical data
- D** They are computationally expensive

#Q. What is the primary goal of data normalization?

- A** To reduce the dimensionality of the dataset
- B** To standardize the range of numerical data
- C** To introduce noise into the dataset
- D** To handle missing values in the dataset

*Transform the data
into Standard Range*

#Q. Which of the following normalization techniques is robust against outliers?

A

Min-Max scaling

B

~~Z-score normalization~~

- less affected by outlier

Min Max

C

Log transformation

D

Unit vector scaling

#Q. When should you use logarithmic transformation for data normalization?

- A** When the data has a skewed distribution
- B** When the data has a normal distribution
- C** When the data has missing values
- D** When the data is categorical

#Q. Which normalization technique is suitable for sparse datasets with mostly zero values?

A

Min-Max scaling

B

Z-score normalization

C

Log transformation

D

Unit ~~vector~~ scaling — normalizes each sample (row)
in to dataset —

[MCQ]



#Q. What does Min-Max scaling do to the data?

- A** Centers the data around zero with unit variance
- B** Standardizes the range of data to a fixed range (e.g., [0, 1])
- C** Transforms the data to a logarithmic scale
- D** Normalizes the data to have unit norm

#Q. What is data generalization in the context of data mining?

- A** Reducing the volume of data for efficient storage
- B** Generating summary or aggregated data from detailed data
- C** Converting continuous data into categorical data
- D** Normalizing the range of numerical data

#Q. Which of the following is an example of data aggregation?

Combining data items

A

Grouping sales transactions by product category

B

Converting temperature readings from Celsius to Fahrenheit

C

Removing outliers from a dataset

D

Scaling numerical data to a fixed range

#Q. Which of the following is NOT a step in data cleaning?

- A** Handling missing values
- B** Removing duplicates
- C** Aggregating data
- D** Correcting inconsistent data

#Q. Which data smoothing technique is used to replace missing values with the most recent non-missing value?

A

Moving average ✗

B

Exponential smoothing

C

✓ Last observation carried forward (LOCF)

D

Savitzky-Golay filter

#Q. Which of the following normalization techniques can handle outliers better?

A Min-Max normalization

B ~~Z-score normalization~~

C Decimal scaling

D Log transformation



THANK - YOU